

An example and results are shown in Figure 8.

Paired t-test

The paired sample t-test, sometimes called the dependent sample t-test, is a statistical procedure used to determine whether the mean difference between two sets of observations is zero. In a paired sample t-test, each subject or entity is measured twice, resulting in pairs of observations.

To perform the paired t-test, choose *Data > Statistics > Paired t-test*.

Data

Variable 1 range: This refers to the range of the first data series to be analysed.

Variable 2 range: This refers to the range of the second data series to be analysed.

Results to: Refers to the top left cell of the range where the test will be displayed.

Grouped by: Selects whether the input data has a *columns* or *rows* layout.

An example and results are given in Figure 9.

F-test

This is any statistical test in which the test statistic has an F-distribution under the null hypothesis. It is most often used when comparing statistical models that have been fitted to a data set, in order to identify the model that best fits the population from which the data is sampled.

To perform the F-test, choose *Data > Statistics > F-test*.

Data

Variable 1 range: This refers to the range of the first data series to be analysed.

Variable 2 range: Refers to the range of the second data series to be analysed.

Results to: Refers to the top left cell of the range where the test will be displayed.

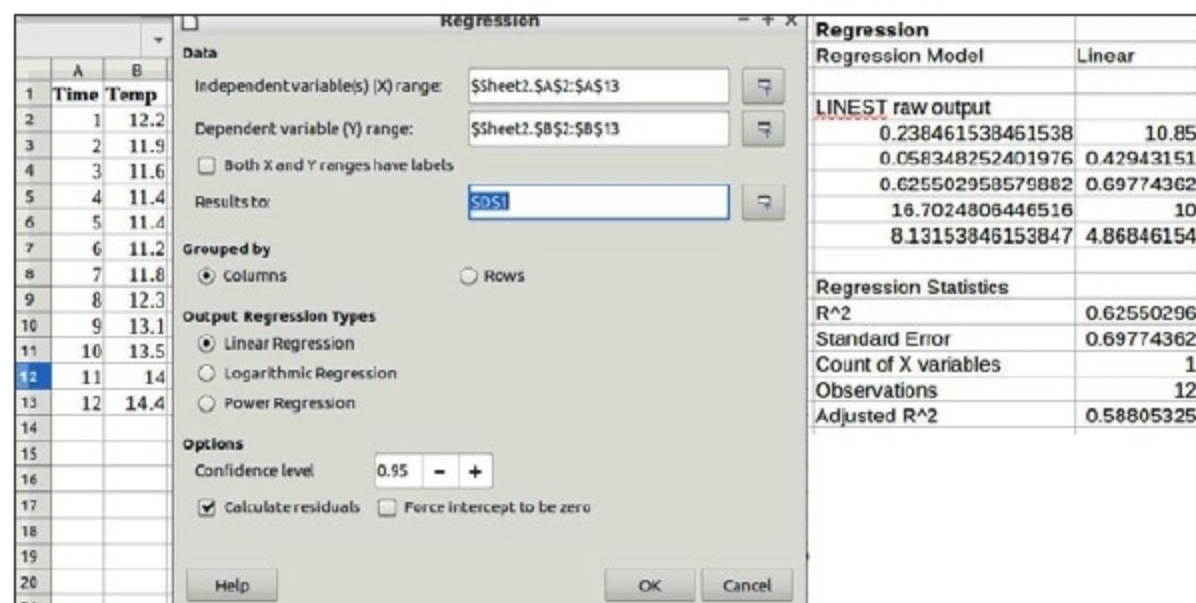


Figure 8: Sample data, options in regression and results of linear regression

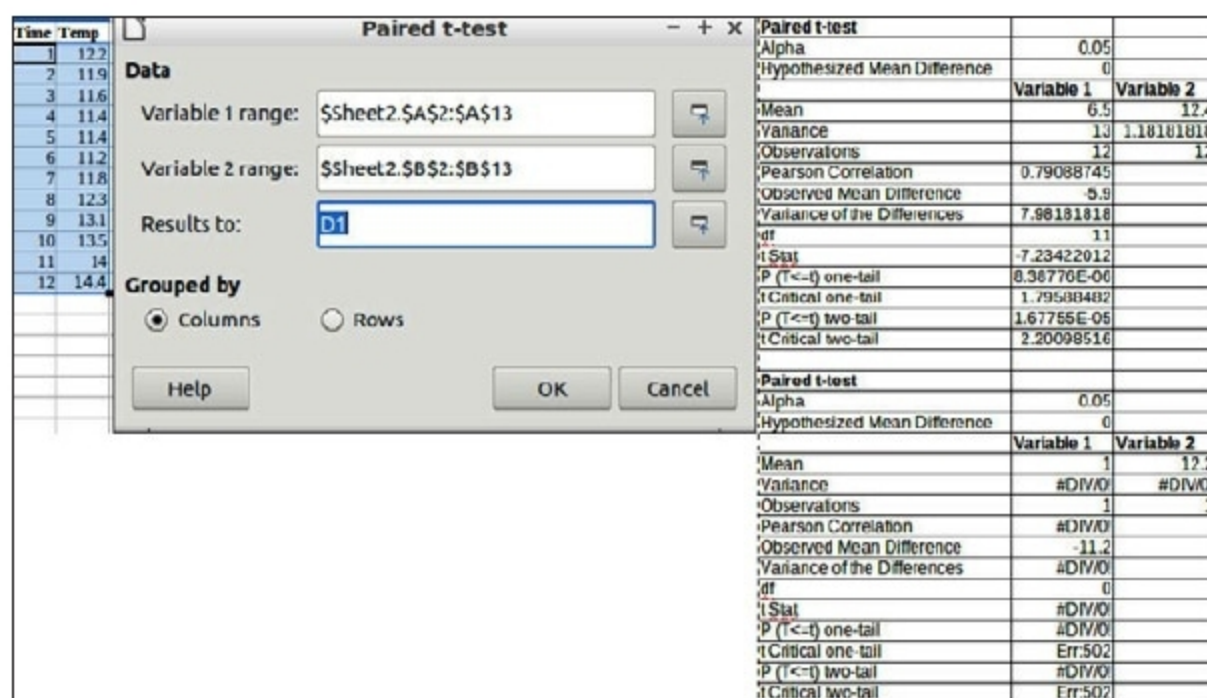


Figure 9: Sample data, options and results of the paired t-test

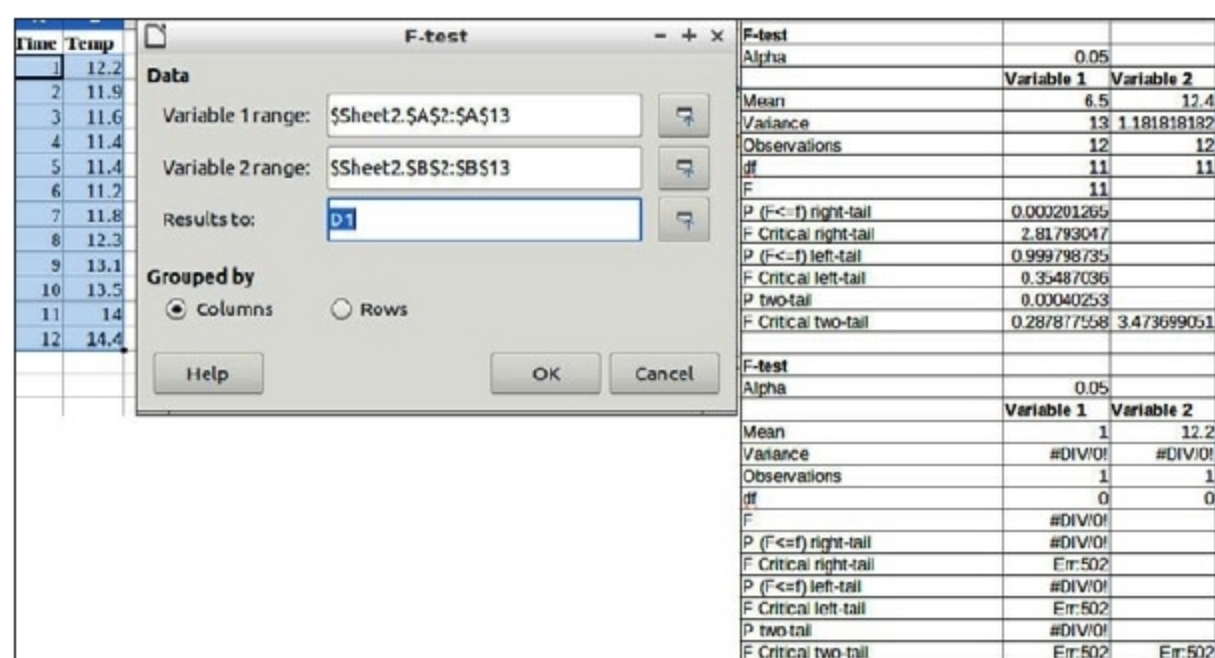


Figure 10: Sample data, F-test options and results

Grouped by: Selects whether the input data has a *columns* or *rows* layout.

An example and results are given in Figure 10.

Z-test

This is a statistical test used to determine whether two population means are different when the variances are known and the sample size is large.